Statistical Consulting Report

Practice 2 Ames Housing Pricing Analysis

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Executive Summary

This analysis provides simple pricing guidelines that can help your agents make quick adjustments when comparing similar homes. On average, a home with an additional 100 square feet of living space can be expected to sell for about $5,600 more, assuming the homes are otherwise similar in quality and location. Neighborhood also has a large impact on value. A home in NridgHt typically sells for about $104,000 more than a comparable home in OldTown. These estimates are intended to serve as practical rules of thumb when evaluating listings or discussing prices with clients. While every home has unique features that affect its value, these guidelines reflect overall market trends and provide a solid starting point for estimating prices.

1. Understanding the Client's Request

The client needs a nontechnical pricing reference that her new agents can use when discussing comparable homes(pg 13). The business goal is to translate the statistical model into two concrete dollar adjustments while accounting for major differences among homes.

The two requested adjustments are: (1) the expected price difference associated with 100 additional square feet of above-ground living area and (2) the expected price premium for NridgHt relative to OldTown. Overall quality is included as an adjustment variable because larger or premium-neighborhood homes may also differ systematically in construction and finish quality.

2. Statistical Questions and Hypotheses

Business Question 1: Living Area

Business question: Holding all else equal, for every additional 100 square feet of living space, how much can the expected sale price increase?

Statistical question: After controlling for overall quality and neighborhood, is above-ground living area associated with sale price, and what is the estimated price difference for 100 additional square feet?

H0: beta_GrLivArea = 0. After adjustment, living area has no association with expected sale price.

HA: beta_GrLivArea != 0. After adjustment, living area has an association with expected sale price.

Business Question 2: Neighborhood Premium

Business question: How much more valuable is a house in NridgHt than a comparable house in OldTown?

Statistical question: After controlling for living area and overall quality, is expected sale price different between NridgHt and OldTown, and what is the estimated adjusted difference?

H0: beta_NridgHt = 0. Comparable homes in NridgHt and OldTown have the same expected sale price.

HA: beta_NridgHt != 0. Comparable homes in NridgHt and OldTown have different expected sale prices.

3. Variable Identification

Variable	Type	Model role	Use in questions
SalePrice	Continuous numeric	Response	Dollar outcome being predicted
GrLivArea	Continuous numeric	Primary predictor	Price change per additional 100 sq ft
OverallQual	Ordinal score (1-10), modeled numeric	Adjustment predictor	Controls for overall material and finish quality
Neighborhood	Nominal categorical (25 levels)	Primary/ adjustment predictor	Compares NridgHt with OldTown and adjusts for location

4. Selection of Statistical Methods

Multiple linear regression was chosen because the response variable, SalePrice, is continuous and the analysis aims to evaluate the effects of both continuous and categorical predictors at the same time. This approach estimates the relationship between each predictor and sale price while holding the other variables constant. The coefficient for GrLivArea represents the expected change in sale price for each additional square foot of living area, after accounting for overall quality and neighborhood. Neighborhood was included as a categorical variable using OldTown as the reference category, allowing the coefficient for NridgHt to estimate the expected price difference between these two neighborhoods. OverallQual was included as a numeric variable (rated from 1 to 10) to account for differences in construction and finish quality. Statistical significance was evaluated using two-sided hypothesis tests and 95% confidence intervals.

Diagnostic plots indicated that the variability of the residuals was not constant across all fitted values (heteroscedasticity). To account for this, HC3 robust standard errors were calculated as a sensitivity analysis. While robust standard errors do not change the estimated regression coefficients, they provide more reliable confidence intervals and significance tests when the assumption of constant error variance is violated.

5. Exploratory Data Analysis

The dataset contained 1,460 sales and 25 neighborhoods. None of the four selected variables had missing values, so all records were used in the analysis.

Measure	SalePrice	GrLivArea	OverallQual
Mean	$180,921	1,515.5	6.10
Standard deviation	$79,443	525.5	1.38
Minimum	$34,900	334	1
25th percentile	$129,975	1,129.5	5
Median	$163,000	1,464	6
75th percentile	$214,000	1,776.8	7
Maximum	$755,000	5,642	10

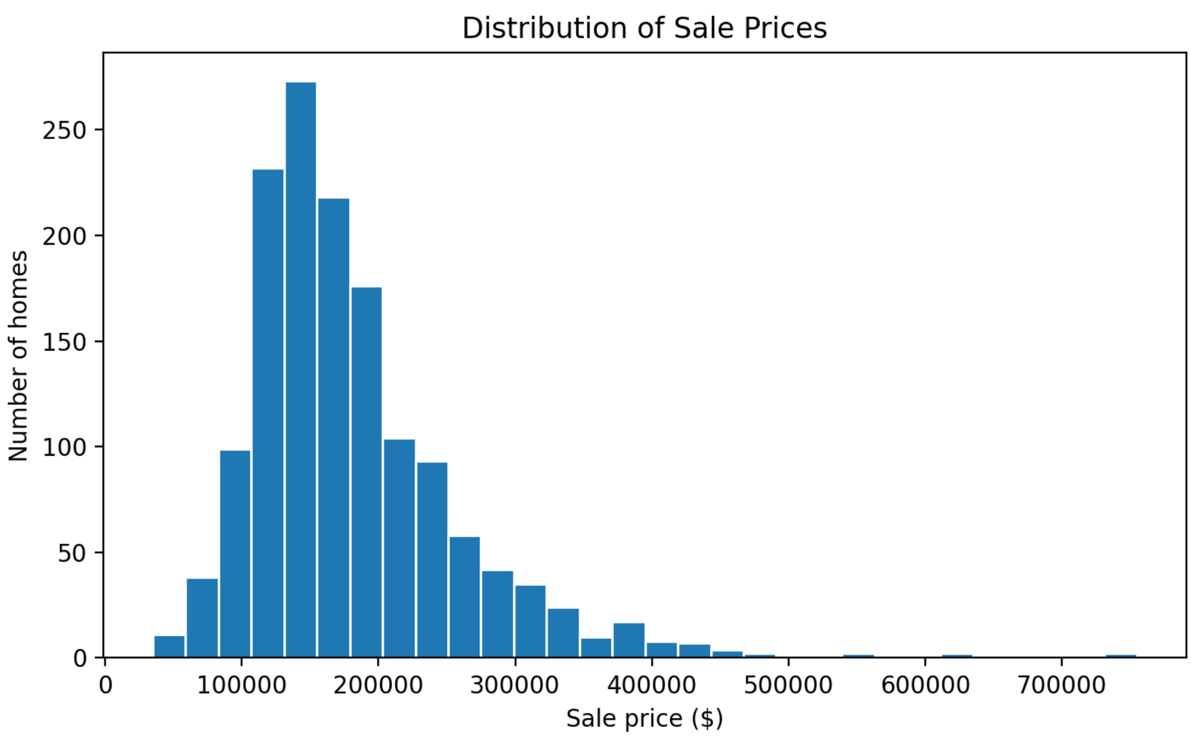

Figure 1. Sale prices are right-skewed, with a small number of very expensive homes.

SalePrice ranged from $34,900 to $755,000, with a median of $163,000. The right tail suggests that very expensive homes may have disproportionate influence on an untransformed model.

Figure 2. Living area and sale price have a strong positive association, with greater spread among larger homes.

The Pearson correlation between SalePrice and GrLivArea was 0.709. The general relationship is positive and approximately linear, although several unusually large or expensive homes are visible and variability increases with fitted price.

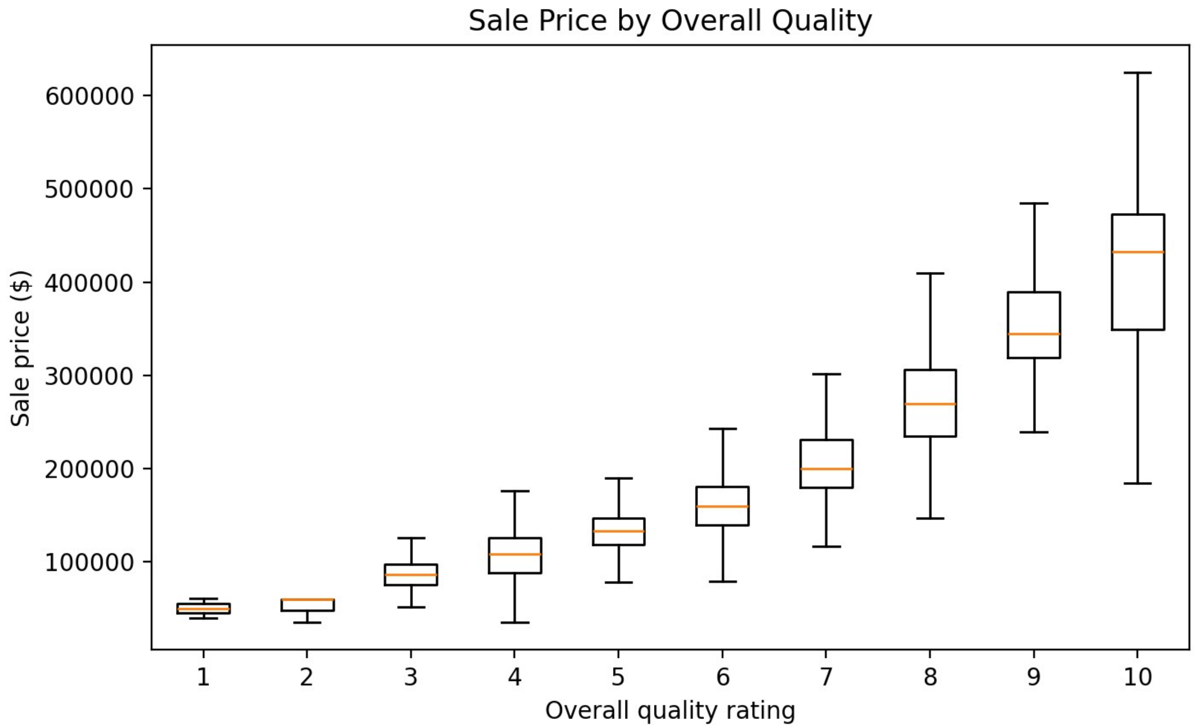

Figure 3. Median sale price rises substantially as overall quality increases.

SalePrice and OverallQual had a correlation of 0.791. The ordered boxplots show a strong monotonic increase, supporting inclusion of quality as an adjustment variable. The correlation between GrLivArea and OverallQual was 0.593; this is moderate rather than so high that the two variables appear redundant.

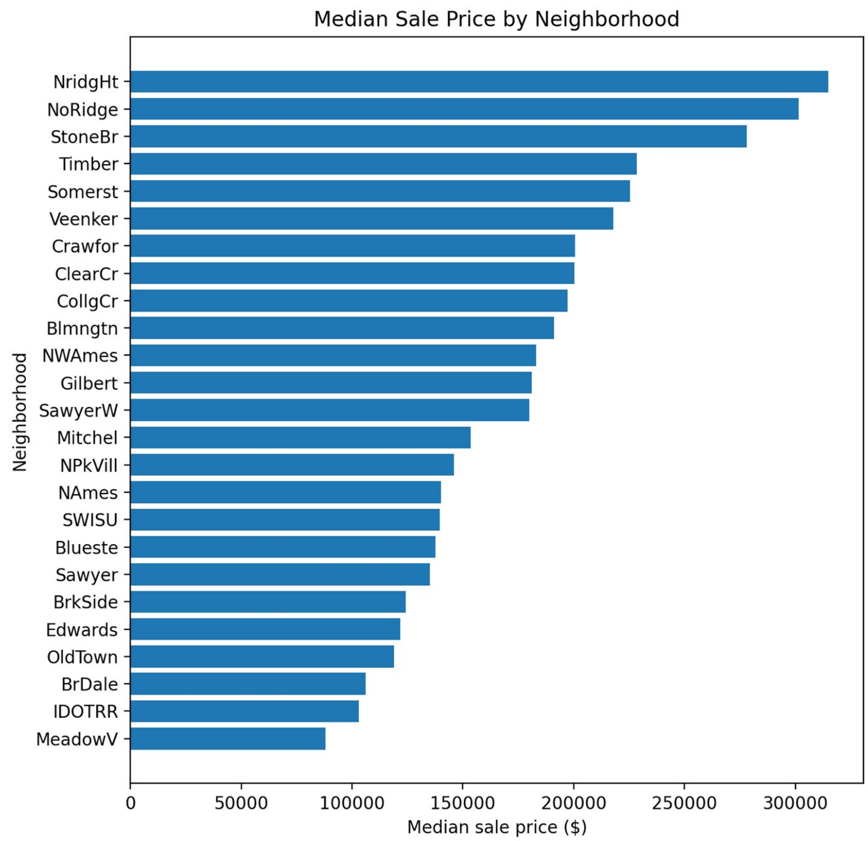

Figure 4. Unadjusted median sale prices differ substantially across neighborhoods.

OldTown included 113 sales with an unadjusted median price of $119,000; NridgHt included 77 sales with an unadjusted median of $315,000. The raw median difference is not the requested answer because the neighborhoods also differ in size and quality. Regression is used to estimate the adjusted difference.

6. Model Specification

The fitted model was:

SalePrice_i = beta_0 + beta_1(GrLivArea_i) + beta_2(OverallQual_i) + sum beta_j(Neighborhood_ji) + epsilon_i

OldTown was the reference neighborhood. Consequently, each neighborhood coefficient is interpreted as the adjusted expected sale-price difference from OldTown. The coefficient for GrLivArea is per one square foot and was multiplied by 100 to answer Brenda's first question.

7. Results

Model statistic	Value
Number of homes	1,460
R-squared	0.787

Model statistic	Value
Adjusted R-squared	0.783
Residual standard error / RMSE	$36,665
Overall F test	$F(26, 1433) = 203.5$, $p < 0.001$

Term	Estimate	95% CI	p-value	Plain-language meaning
GrLivArea (1 sq ft)	$55.56	$50.66 to $60.47	< 0.001	Adjusted change in expected price per additional sq ft
GrLivArea (100 sq ft)	$5,556	$5,066 to $6,047	< 0.001	Adjusted change for 100 additional sq ft
OverallQual (1 point)	$20,951	$18,671 to $23,231	< 0.001	Adjusted change for a one-point higher quality rating
NridgHt vs OldTown	$103,669	$91,481 to $115,856	< 0.001	Adjusted neighborhood premium for NridgHt

Holding overall quality and neighborhood constant, the model estimates $55.56 per additional square foot, or $5,556 per 100 square feet. Holding size and quality constant, NridgHt is estimated to have a $103,669 premium relative to OldTown. Both coefficients are statistically significant at $p < 0.001$. A one-point increase in OverallQual is associated with an estimated $20,951 increase in expected sale price, after the other variables are held constant.

The model R-squared of 0.787 means that living area, overall quality, and neighborhood together explain approximately 78.7% of the observed variation in SalePrice. The RMSE of about $36,665 indicates that individual sale prices commonly differ from fitted values by tens of thousands of dollars, reinforcing that the cheat-sheet values are average adjustments rather than precise predictions for a specific property.

8. Model Validation

Linearity and Functional Form

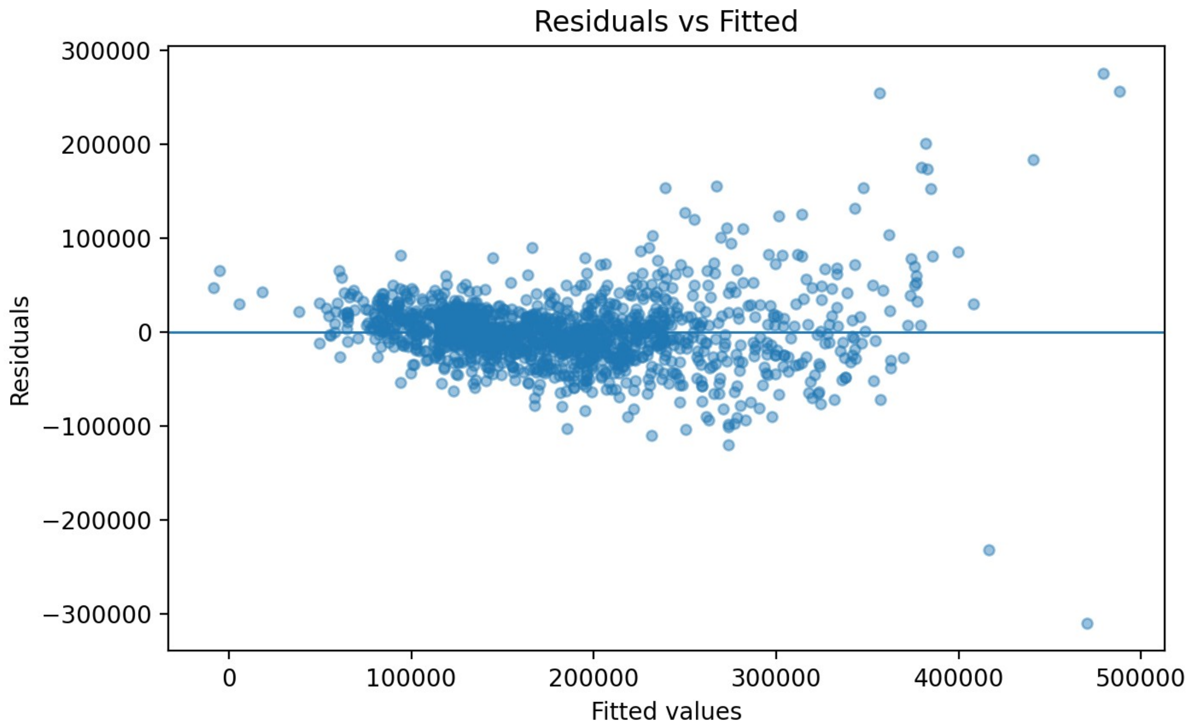

Figure 5. Residuals versus fitted values show increasing spread and several large residuals.

The residual plot is centered around zero, but the spread grows as fitted price increases and several observations have large residuals. The linear mean structure is useful for a simple client rule, but the plot suggests that a transformation, nonlinear terms, or interactions may improve fit for expensive homes.

Normality of Residuals

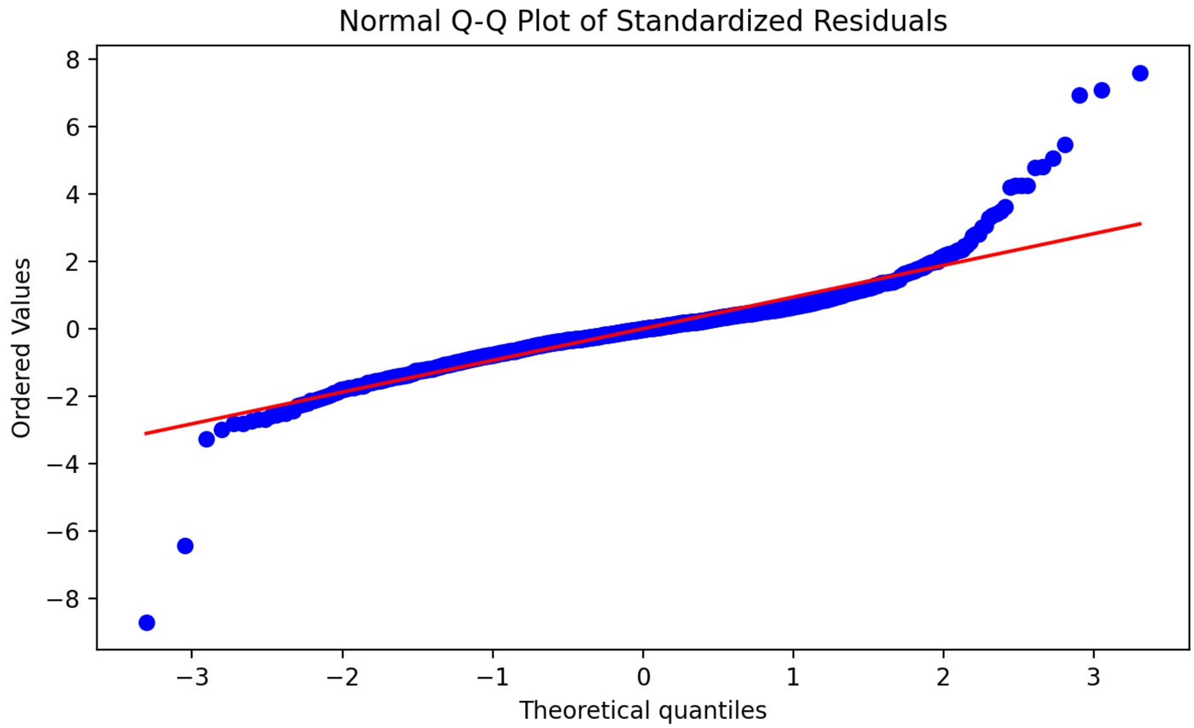

Figure 6. The Q-Q plot shows substantial tail departures from normality.

The Shapiro-Wilk test rejected normality ($W = 0.879$, $p < 0.001$). With 1,460 observations, formal tests are sensitive to small departures, but the Q-Q plot also shows meaningful heavy tails. Coefficient estimates remain useful, but conventional confidence intervals should be interpreted cautiously.

Constant Variance

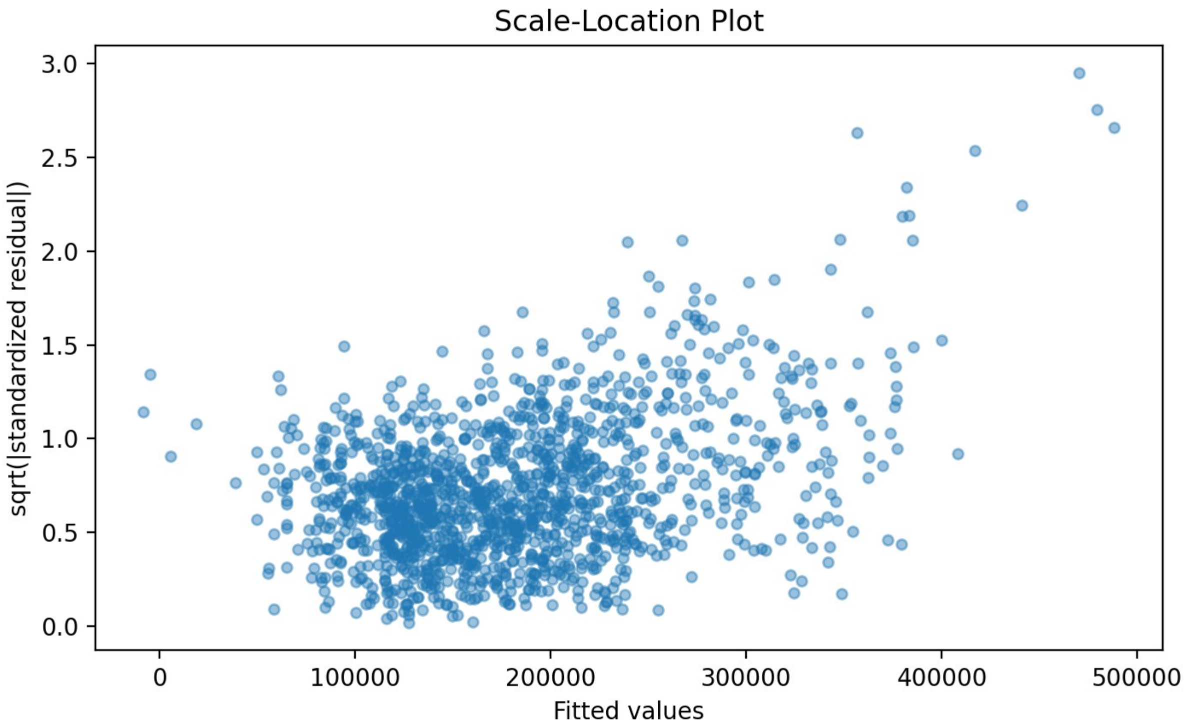

Figure 7. Residual variability increases with fitted sale price.

The Breusch-Pagan test rejected constant variance (chi-square = 329.0, $p < 0.001$). Therefore, HC3 robust confidence intervals were examined. The robust 95% interval was $4,251 to $6,862 for 100 square feet and $84,885 to $122,452 for NridgHt versus OldTown. Both still exclude zero, so the primary conclusions remain the same

Multicollinearity

GrLivArea and OverallQual showed a moderate positive correlation (0.593), indicating that larger homes tend to have higher quality ratings, but not to the extent that either variable should be excluded from the model. Because Neighborhood is a categorical variable represented by multiple indicator variables, generalized variance inflation factors (GVIFs) were used to assess multicollinearity. The results indicated that multicollinearity was not a concern, as the regression coefficients remained stable and both GrLivArea and OverallQual continued to be strong predictors of sale price. While some neighborhoods with relatively few observations had less precise estimates, this did not affect the interpretation of the primary comparisons used to answer the client's questions.

Influential Observations

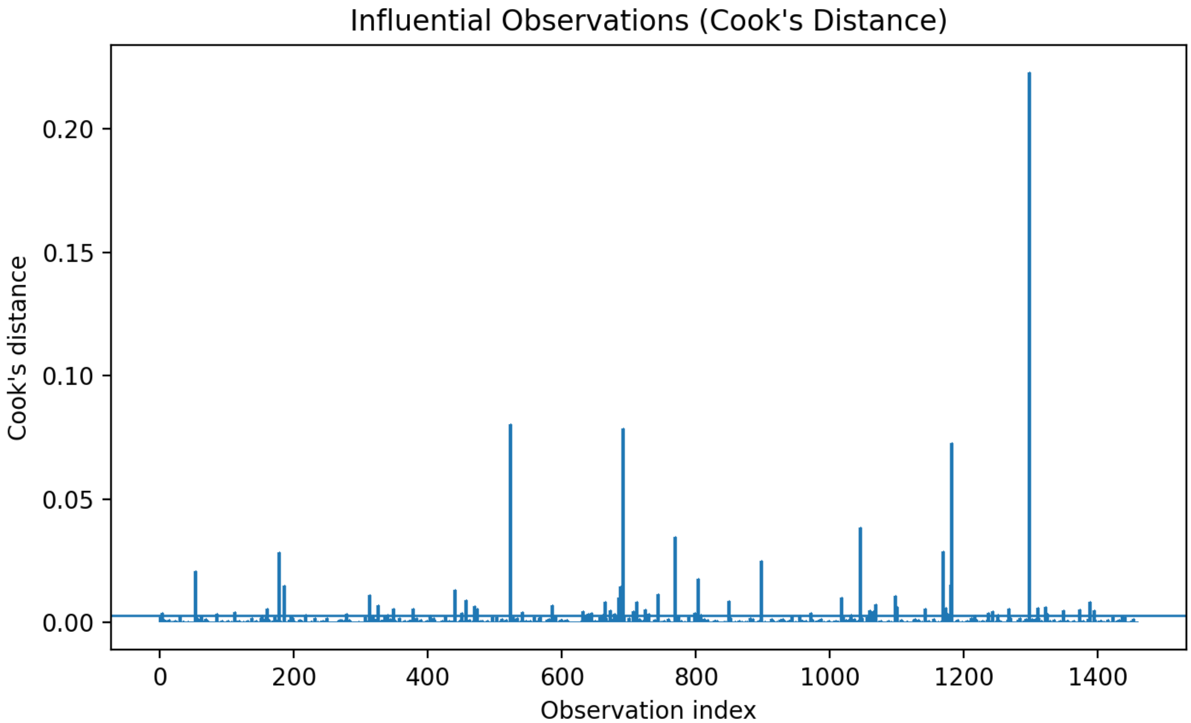

Figure 8. Cook's distance; the horizontal line is the common 4/n screening threshold (0.0027).

Cook's distance was used to identify observations that may have an unusually large influence on the regression results. Using a common screening guideline, 75 homes were identified as potentially influential, and the largest Cook's distance observed was 0.223. These observations are not automatically considered problematic and should not be removed without further investigation. Many likely represent legitimate high-value or otherwise unusual properties rather than data errors. If this model were to be used for formal property appraisal or other high-stakes decisions, a sensitivity analysis—such as refitting the model without these influential observations or modeling luxury homes separately—would help determine whether the conclusions remain consistent.

Independence

The **Durbin–Watson statistic** was **1.949**, indicating no evidence of substantial autocorrelation in the model residuals. Because the data consist of individual home sales collected at a single point in time rather than repeated measurements over time, the assumption of independent observations is generally reasonable. However, the dataset does not provide information on potential clustering, such as homes sold by the same seller or agent or repeated sales of the same property. Additionally, nearby homes may share similar characteristics that could introduce some spatial dependence among the residuals9. Answers to Client Questions

Client question	Direct answer
What is the expected change for 100 additional square feet?	Approximately $5,556 higher sale price, holding overall quality and neighborhood constant (conventional 95% CI: $5,066 to $6,047; HC3 robust CI: $4,251 to $6,862).
What is the premium for NridgHt versus OldTown?	Approximately $103,669 higher sale price for a home with the same living area and overall quality (conventional 95% CI: $91,481 to $115,856; HC3 robust CI: $84,885 to $122,452).

10. Recommendations

These pricing adjustments can be used as a practical starting point when comparing similar homes. As a general guideline, adding 100 square feet of living space increases a home's value by about $5,600, provided the homes are similar in quality and located in comparable neighborhoods. Likewise, a home in NridgHt can be expected to sell for about $104,000 more than a similar home in OldTown. Because every property is unique, these estimates should be viewed as approximate rather than exact values. Factors such as the home's condition, lot size, age, renovations, number of bedrooms and bathrooms, garage, and current market conditions can all affect the final sale price. Homes that are unusually large, expensive, or otherwise unique should be evaluated individually instead of relying solely on these guidelines.

11. Limitations

The pricing guidelines in this report are intended to provide a helpful starting point, not an exact valuation for every home. They are based on historical home sales in Ames and consider only living area, overall quality, and neighborhood. Other important factors—such as the home's condition, lot size, age, renovations, number of bedrooms and bathrooms, garage, and current market conditions—can also have a significant impact on sale price but were not included in this analysis. In addition, the relationship between these factors and price may not be the same for every type of home or neighborhood, particularly for luxury or unique properties. Because the data come from one housing market and one period of time, these estimates may not apply as accurately to other locations or to today's market. For these reasons, the pricing adjustments should be used as general guidelines and should always be combined with professional judgment and a thorough review of comparable properties.

12. Future Work

This model could be improved by including additional information about each home, such as its age, condition, lot size, number of bedrooms and bathrooms, garage size, and when it was sold. It may also be beneficial to allow the relationship between home size, quality, and neighborhood to vary rather than assuming they affect every home in the same way. Future versions of the model should also be tested on new data to evaluate how accurately they predict the prices of homes that were not used to build the model. Finally, the model could be expanded into a simple pricing calculator that provides both an estimated sale price and a reasonable range of likely values for an individual home.

Appendix

The accompanying file Ames_Housing_Complete_Analysis.R contains the complete reproducible workflow: data import, cleaning, EDA, figures, model fitting, coefficient extraction, confidence intervals, robust inference, diagnostic tests, influence analysis, and export of tables and plots.

Primary model formula: SalePrice ~ GrLivArea + OverallQual + Neighborhood, with OldTown as the reference category.

Software: R. The script uses tidyverse, broom, car, lmtest, sandwich, performance, and patchwork. Package installation commands are included as comments.

Brenda's Home Pricing Cheat Sheet

Quick Pricing Adjustments for Similar Homes

Living Space

Rule of Thumb:
For homes that are similar in quality and neighborhood, every additional 100 square feet adds about $5,600 to the home's value.

Difference in Living Area	Estimated Price Adjustment
100 sq ft	+$5,600
200 sq ft	+$11,100
500 sq ft	+$27,800

Neighborhood Matters

A home in NridgHt typically sells for about $104,000 more than a similar home in OldTown. When using an OldTown comparable to estimate a NridgHt listing, increase the estimated value by roughly $104,000.

Quick Example

Feature	Home A	Home B
Neighborhood	OldTown	NridgHt

Living Area	1,800 sq ft	2,000 sq ft
Overall Quality	Similar	Similar

Estimated adjustments:
• 200 additional square feet: +$11,100
• NridgHt neighborhood: +$104,000

Estimated difference: About $115,000

Keep in Mind

These adjustments work best when homes are otherwise similar in overall quality, condition, age, and amenities. Use them as a starting point alongside comparable sales and professional judgment.

Bottom Line

• +100 sq ft → Approximately +$5,600
• NridgHt vs. OldTown → Approximately +$104,000

These practical market-based adjustments can help you make faster, more consistent pricing decisions.